KYURI PARK

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EDUCATION

University of Amsterdam - Computational Science Lab, Informatics Institute Aug 2023-Present
- PhD Candidate; expected completion August 2027.
- Supervisors: Vítor V. Vasconcelos and Mike Lees.

Visiting Scholar, Stanford University - Stanford Doerr School of Sustainability Spring 2026
- Hosted by Sara Constantino; research on climate-attitude networks, belief systems, social norms, and exploratory causal discovery.

Complex Systems Summer School, Santa Fe Institute Summer 2026
- Selected participant; interdisciplinary training and collaborative modeling projects in complex systems.

MSc, Methodology and Statistics, Utrecht University Jun 2023
- Cum laude, GPA 9.04/10.00.
- Supervisors: Oisín Ryan and Lourens J. Waldorp.

BSc, Psychology (Psychological Methods), University of Amsterdam Jun 2021
- Cum laude, GPA 9.05/10.00; Honors Programme.
- Supervisors: Claudia van Borkulo and Denny Borsboom.

BSc, Business Management, University of Houston May 2017
- Summa cum laude, GPA 3.94/4.00; specialization in Finance Management and Accounting.
- Exchange semester at Mount Royal University, Calgary, Canada.

RESEARCH INTERESTS

Complex systems | Causal discovery | Dynamical systems | Network analysis | Computational modeling | Multiscale dynamics | Mental health | Climate beliefs, behavior, and adaptation

PEER-REVIEWED PUBLICATIONS

1. **Park, K.**, Waldorp, L. J., & Vasconcelos, V. V. (2026). A mechanistic model of symptom dynamics: Implications for statistical network analyses. *Multivariate Behavioral Research*. Advance online publication. doi:10.1080/00273171.2026.2710730.
2. **Park, K.**, Vasconcelos, V. V., & Lees, M. (2026). A tutorial on causal network simulation and exploration using the *causalnet* R package. *Behavior Research Methods*, 58, 204. doi:10.3758/s13428-026-03030-z.
3. **Park, K.**, Li, X., Waldorp, L. J., Lees, M., & Vasconcelos, V. V. (2026). The role of feedback loops in dynamical symptom networks. *Scientific Reports*, 16, 11273. doi:10.1038/s41598-026-38747-6.
4. **Park, K.**, Elsenburg, L. K., Nicolaou, M., Stronks, K., & Vasconcelos, V. V. (2026). Connecting precariousness and depression: From causal discovery to intervention simulation. *SSM - Mental Health*, 9, 100637. doi:10.1016/j.ssmmh.2026.100637.
5. Elsenburg, L. K., Stronks, K., Galenkamp, H., Lakerveld, J., Lok, A., **Park, K.**, Vasconcelos, V. V., & Nicolaou, M. (2026). Precariousness and depressed mood: A network analysis in the multi-ethnic HELIUS study. *Social Psychiatry and Psychiatric Epidemiology*. doi:10.1007/s00127-026-03072-w.
6. **Park, K.**, Waldorp, L. J., & Ryan, O. (2024). Discovering cyclic causal models in psychological research. *advances.in/psychology*, 2(1), e72425. doi:10.56296/aip00012.

Preprint
- Lee, S. Y., Hwang, H., Kim, T., Kim, Y., **Park, K.**, Yoo, J., Borsboom, D., & Shin, K. (2025). Emergence of psychopathological computations in large language models. arXiv. doi:10.48550/arXiv.2504.08016.
- **Park, K.**, Borsboom, D., Lees, M., Elsenburg, L., Lunansky, G., Stronks, K., … Vasconcelos, V. (2026). Rethinking Group Differences in Psychopathology Networks: A Slow–Fast Perspective on Context and Symptom Activation. osf.io/preprints/psyarxiv/u8wf5_v1

Selected research in progress
- Time-scale separation in dynamical models of psychological processes
- Computational modeling of executive-function engagement across development.
- Comparative climate-attitude networks and exploratory causal discovery across Japan, Europe, and the United States.
- Institutionalized category convergence in heterogeneous rate-distortion agent-based models.
- Post-disaster population displacement, migration, and urban adaptation.

GRANTS, FELLOWSHIPS, AND AWARDS

- Van der Gaag Grant, Royal Netherlands Academy of Arts and Sciences (KNAW) - PI, 2026-2027.
- Bridge Grant, Complex Systems Society, 2026.
- Energy Transition through the Lens of the SDGs (ENLENS) Research Grant, 2025-2027.
- Travel Award, International Meeting of the Psychometric Society, 2025 and 2026.
- Amsterdam University Fund - PhD Grant, 2025.
- Utrecht Excellence Scholarship, Utrecht University, 2021-2023.
- Scholarship Excellence Award, University of Houston, 2017
- UH Academic Excellence Scholarship, University of Houston, 2014-2017.

SELECTED PRESENTATIONS

- Park, K. (Jul 2026). A slow-fast perspective on between-group differences in symptom networks. International Meeting of the Psychometric Society, Seoul, South Korea.
- Park, K. (Jul 2025). Computational modeling of symptom dynamics and their implications for statistical networks. International Conference on Computational Science, Singapore.
- Park, K. (Jul 2025). Unraveling symptom dynamics: A mechanistic approach to feedback loops and causal discovery. International Meeting of the Psychometric Society, Minneapolis, MN.

INVITED TALKS AND SEMINARS

- Park, K. (May 2026). Executive function engagement as a slow-fast stochastic dynamical system. Center for Mind and Brain, University of California, Davis, Davis, CA, USA.
- Park, K. (Apr 2026). Complex systems approaches to mental health and climate-change attitudes. Biohealth Convergence and Open Sharing System (COSS), South Korea.
- Park, K. (Feb 2026). Slow-fast coupling: Curie-Weiss symptoms under context dynamics. 3rd Joint Kobe University-UvA Workshop on Computational Social Science and Intelligent Systems, Amsterdam.
- Park, K. (Jan 2026). Structural risk shifts symptom thresholds without altered symptom coupling: A slow-fast perspective. Theory Methods Lab Seminar, Amsterdam.
- Park, K. (Mar 2025). Computational approaches to mental health at the symptom level. 2nd Joint Kobe University-UvA Workshop on Computational Social Science and Intelligent Systems, Amsterdam.

RESEARCH EXPERIENCE

University of Amsterdam - Computational Science Lab, Informatics Institute Aug 2023-Present

- Develop causal and dynamical models of complex human systems, with applications in mental health, climate attitudes, belief dynamics, and urban adaptation.
- Combine causal discovery, network analysis, agent-based modeling, and simulation to study feedback processes, structural uncertainty, and intervention effects.
- Lead developer of `causalnet`, an open-source R package for generating alternative directed structures from a network skeleton, including networks with cycles, summarizing their properties, and simulating nonlinear or user-defined dynamics.

Utrecht University Jun 2021-Aug 2023

- Enhanced the R package `mHMMbayes` (Multilevel Hidden Markov Models Using Bayesian Estimation) and developed tutorial materials.
- Contributed to work on the `mice` R package and developed a synthetic-data approach for evaluating imputation-model validity with Gerko Vink.
- Developed procedures for producing reproducible synthetic-data extracts from Statistics Netherlands data and co-developed the `synthpop.extract` R package with Erik-Jan van Kesteren.
- Translated a Bayesian piecewise-linear regression implementation for dynamic network construction from MATLAB to R.
- Applied and compared cyclic causal-discovery algorithms (CCD, FCI, CCI) for psychological data.
- Coded methodological characteristics of psychological network studies in NVivo with Noémi Schuurman.

University of Amsterdam - Psychological Methods Nov 2019-Jan 2021

- Conducted a meta-analysis of predictors of science skepticism, developed JASP tutorials, and modeled networks containing variables operating at different timescales using Ising and mixed graphical models.

TEACHING AND STUDENT MENTORING

University of Amsterdam

- Teaching Assistant, *Complexity: Can It Be Simplified?* (BSc Honors/MSc), Vítor V. Vasconcelos, 2023–2025.
- Teaching Assistant, *Causality* (MSc Artificial Intelligence), Sara Magliacane, 2024–2025.
- Teaching Assistant, *Modeling System Dynamics* (MSc Information Studies), Debraj Roy, 2023–2024.
- Teaching Assistant, *AI and Psychology* (BSc Psychology), Han van der Maas, 2021.

Utrecht University

- Developed practical materials for Python programming, Bayesian networks, relational-event models, missing data, imputation, and text mining courses; provided R programming support to sociology graduate students.

University of Pennsylvania

- Teaching Assistant, Social Economics: An Economic Perspective on Human Behavior, Femida Handy, Summer 2016.

MSc Student Supervision

- **Henry Zwart**, MSc in Computational Science (2026). Thesis: *Interventions on Asymmetric Belief Networks.*
- **Paraskevas Leivadaros**, MSc in Data Science (2025). Thesis: *Elections and climate attitudes: how do people's views on climate change and related policies change during an election?*
- **Gabriela Torres**, MSc in Data Science (2025). Thesis: *The Impact of COVID-19 Interventions and Extreme Weather Harm on Support for Climate Policy*.
- **Xinhai Li**, MSc in Computational Science (2024). Thesis: *Feedback Loops in the Dynamics of Major Depressive Disorder*.

ADDITIONAL EXPERIENCE

SERVICE AND LEADERSHIP

University of Amsterdam 2018-2021

- International Student Ambassador and member of the Psychological Methods Student Faction; represented students in programme meetings and organized academic, orientation, and outreach events.

Utrecht University - *Human Data Science Group* 2021-2023

- Organized research and data-science reading groups, helped coordinate Data Donation Day, and maintained the group website.

PROFESSIONAL EXPERIENCE

Emirates Airline, Dubai, UAE May 2017-Jul 2018

- Cabin crew

Amerril Energy LLC, Houston, TX, USA Jun 2016-Apr 2017

- Accounting assistant

VOLUNTEER EXPERIENCE

World Vision 2011-Present

- Sponsor since 2011; previously translated sponsor correspondence between Korean and English.

Stray Dogs Center (SDC), Dubai, UAE Jul 2017-Jun 2018

- Volunteered at the shelter and supported fundraising for veterinary expenses.

METHODS, PROGRAMMING, AND LANGUAGES

Methods: Causal discovery, network analysis, dynamical-systems modeling, stochastic differential equations, agent-based modeling, simulation studies, psychometric modeling, and Bayesian and multilevel modeling.

Programming: R, Python, MATLAB; JavaScript (learning), C++ (learning)
Languages: Korean (native), English (fluent), Dutch (B1).